Problem 1

Object A has 5.7 times the mass and 1.4 times the volume of object B. If the density of object B is 731 kg/m^3, find the density of object A in the same units.

ρ = _____ kg/m^3

Problem 2

A hollow cylindrical pipe is made from material 2 listed in the table below. The pipe is 3.8 m long, has an outside diameter of 3.7 cm, and an inside diameter of 2.1 cm. Calculate the pipe's weight in lbs. (1 lb = 4.4482 N)

	Material	Density (kg/m^3)
1	Copper	8.9×10^3
2	Iron	7.8×10^3
3	Lead	11.3×10^3

w = _____lbs

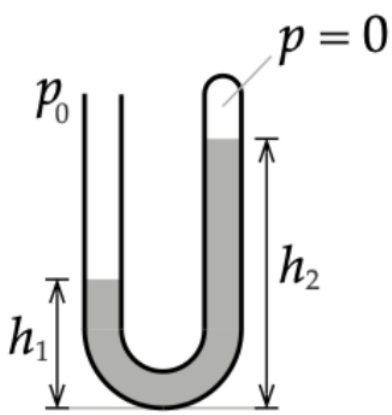
Problem 3

A cube of mass 24 kg and side 29 cm is placed on a smooth, flat surface in a vacuum on a planet where the acceleration of gravity is 5.6 m/s^2. Calculate the pressure on the surface beneath the cube.

p =_____Pa

Problem 4

Reading the scale on the mercury barometer in the figure, a researcher on Planet X finds that h_1 = 7.5 cm and h_2 = 60.2 cm. Calculate the ambient air pressure in kPa. Density of mercury: 13,534 kg/m^3. 1 kPa = 1000 Pa. The acceleration of gravity on Planet X is 7.2 m/s^2.



$p_0 =$ _____ kPa

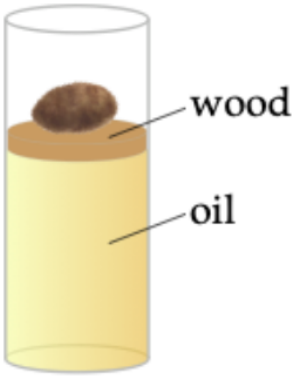
Problem 5

What is the pressure at a depth of 6.09 m at a freshwater lake, if the atmospheric pressure is 96,872 Pa at the lake's surface? The density of freshwater is 1000 kg/m^3.

p = _____Pa

Problem 6

The figure shows a glass cylinder of radius 6.3 cm, partially filled with oil of density 0.81×10^3 kg/m^3. Floating freely on top with no oil leaking around the sides, a 2.2-kg wooden disk supports a 0.4-kg potato. Calculate the gauge pressure in psi at 12.1 cm below the surface of the oil. (1 psi = 6.895 kPa)



p = _____psi (Note the units!)

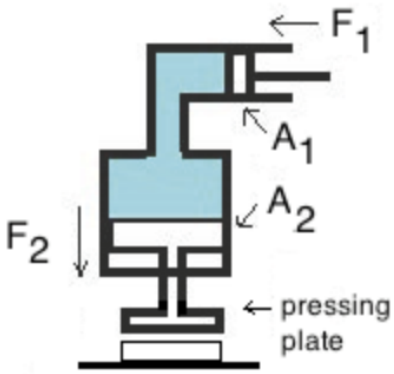
Problem 7

A hydraulic jack has input piston with radius 0.28 m and output piston with radius 1.62 m. If the input force is 388 N, calculate the output force in kN.

F = _____kN

Problem 8

A small hydraulic press has input piston of radius 6 cm and output piston of radius 24 cm. The output piston is connected to a pressing plate of area 9.5 cm^2. If the input force is F_1 = 246 N, calculate the compressive stress in kPa exerted by the press.



p = _____kPa

Problem 9

A ball of volume 12 m^3 is placed in a container of fluid of density 783 kg/m^3. Calculate the density of the object, if it floats and 7.9 m^3 of it is submerged in the fluid.

ρ = ____ kg/m^3

Problem 10

A solid wooden raft having density 511 kg/m^3 and dimensions 3.5 m by 2.5 m by 0.4 m, where the latter is the thickness, is floating in water. How much mass can be loaded on the raft before 83% of its volume is submerged?

m = _____kg

Problem 11

A pipe of diameter 0.26 m increases to a diameter of 0.84 m. If the fluid is traveling at 6.4 m/s in the narrower part of the pipe, what is the fluid speed in the wider part of the pipe?

v = _____m/s

Problem 12

How many minutes does it take to fill a 39-gallon bathtub if the faucet opening has an area of 1.5 cm^2 and water flows out of it at 3.5 m/s? (1 gal = 3.785 L = 3.785×10^{-3} m^3.)

t = _____ min

Problem 13

A chamber 39 m beneath the Earth contains hot water with a density of 1000 kg/m^3 at a pressure of 796,015 Pa. A narrowing channel leads upwards to a circular vent at the surface having negligible area compared to that of the chamber. At what speed is hot water ejected from the vent? Atmospheric pressure is 1.01 x 10^5 Pa

v = _____m/s

Problem 14

A pipe of constant radius carries water horizontally at a pressure 177 kPa and speed 3 m/s when it curves upward, rising 2.9 m before returning to horizontal. Calculate the pressure p in kPa at the new height. The density of water is 1000 kg/m^3.

p = _____ kPa

Problem 15

Two solid metal balls all have the same radius. One is made of aluminum, the other of gold, and their densities are known. If the balls are resting at the bottom of a pool of water, which ball has the largest buoyant force?

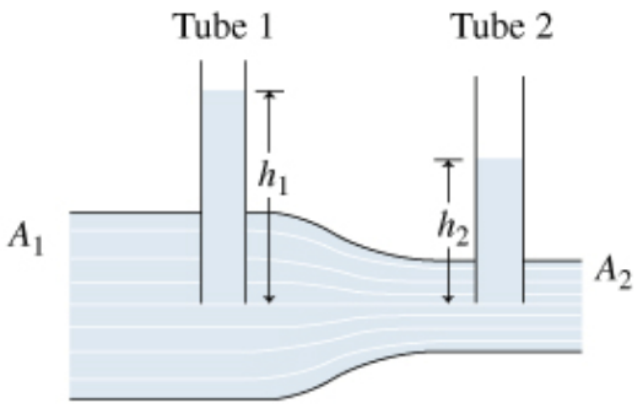
- The gold ball has the larger buoyant force.
- The buoyant forces on the balls are the same
- The aluminum ball has the larger buyant force
- More information is required

Problem 16

The figure below shows an ideal liquid in a Venturi pipe.

The liquid's speed of is greater at the location of tube 1 .

The pressure is greater at the location of tube 2 .



Answer 1:

the location of tube 1

the location of tube 2

Answer 2:

the location of tube 2

the location of tube 1